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MILITARY SPECIFICATION

INSIGNIA, BRANCH OF SERVICE; GARRISON HAT; CAP; AND SERVICE CAP OFFICERS), OFFICERS, OFFICER CANDIDATES AND AVIATION CADETS, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers branch of service insignia, hat insignia, garrison cap insignia, and service cap insignia used by officers, officer candidates and aviation cadets.

1.2. Classification. The insignia covered by this specification shall be as specified herein and on the applicable Military Specification Sheets (see 2.1, 3.1, 6.3 and Supplement 1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions, and any pertinent data which may be used in improving this document should be addressed to: Director, The Institute of Heraldry, US Army, Cameron Station, Alexandria, Va 22304-5050, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8455

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SPECIFICATIONS

FEDERAL

QQ-N-290 - Nickel Plating
QQ-Z-363 - Zinc Base Alloy, Die Castings
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-566 - Boxes, Folded, Paperboard
PPP-B-676 - Boxes, Setup
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-F-495- Finish, Chemical, Black for Copper Alloys

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods
FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

SPECIFICATION SHEETS

(See supplement 1 for list of associated specifications).

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

THE INSTITUTE OF HERALDRY

B-13-12 - Attaching Devices, Hook and Clutch Type
B-1 (Series) - Insignia, Cap, Officer

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B-2 (Series) - Branch of Service, Officer

STANDARD COLOR CHIP SET

The Institute of Heraldry, U.S. Army Standard Color Chip Set

(The standard color chips may be obtained from the procuring activity for Government procurements).

(Copies of drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

(Figure 1 is a miniature copy of drawing B-13-12 and is for information only).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

ASTM-D-3951 - Standard Practice for Commercial Packaging.
ASTM-B-487 - Metal and Oxide Coating Thickness by Microsopical Examination of A Cross Section, Measurement of

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103) and (Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Standard sample. Unless otherwise specified (see 6.3), the contracting officer will furnish a standard sample or approved guide sample which shall be used for matching color and finish only. Variations from this specification may appear in the standard sample or approved guide sample; however, in such cases the specification shall govern.

3.3 First article. Unless otherwise specified (see 6.4), before production is commenced, a sample of the finished insignia covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for inspection and tests as specified in 4.4. The approval of the first article does not relieve the contractor of responsibility for compliance with all applicable provisions of this specification. The first article shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production items.

3.4 Materials. Materials shall conform to the referenced specifications and the requirements specified herein.

3.4.1 Copper base alloy. The copper base alloy shall be roll polished, free from pits, scale (including red oxide), dents, nicks, cracks, scratches, segragations, and foreign inclusions that will not be removed in later processing. When tested as specified in 4.7.1, the chemical composition of the applicable copper base alloy shall be as specified in table I.

TABLE I Chemical composition of copper base alloys.

Alloy	Copper	Nickel 1/	Lead (maximum)	Iron (maximum)	Manganese (maximum)	Zinc	Others (maximum)
Red Brass	84.0 - 86.0		.05	.05		Remainder	.15
Gilding Metal	94.0 - 96.0		.03	.05		Remainder	.13
Free cutting brass	60.0 - 63.0		2.50 3.70			Remainder	
Nickel silver	63.0- 66.5	16.5-19.5	.10	.25	.50	Remainder	

1/ Cobalt counting as nickel.

3.4.2 Solder.

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3.4.2.1 Hard solder. Hard solder shall be a silver alloy solder having a melting point of not less than 1075 degrees F.

3.4.2.2 Soft solder. Soft solder shall be a tin-lead alloy having a melting point of not less that 375 degrees F.

3.4.3 Lacquer. The lacquer shall be a pale clear synthetic lacquer. The use of the pigmented lacquer shall not be permitted.

3.4.4 Gold for plating. Gold for plating shall be 24 karat.

3.4.5 Silver.

3.4.5.1 Sterling Silver. Sterling silver shall contain not less than 92.5 percent pure silver.

3.4.5.2 Silver for plating. Silver for plating shall contain not less than 99.5 percent pure silver.

3.4.5.3 Silver filled material. When silver filled material is specified, the silver alloy shall not be less than 92.5 percent pure silver. The base of the silver filled material shall be any copper base alloy having a specific density of not less than 96 percent pure copper.

3.4.6 Enamel.

3.4.6.1 Hard Enamel. Hard enamel shall be a hard glass type vitreous enamel fused with metallic oxides to produce the desired color, shade and opacity or translucence.

3.4.6.2 Soft enamel. Soft enamel shall be a baking enamel pigmented to produce the required color or shade.

3.4.6.2.1 Subdued insignia. The enamel for subdued insignia shall be a flat synthetic, baking enamel pigmented to black shade 37038 established in FED-STD-595.

3.4.7 Zinc aluminum alloy. The alloy used for die casting shall conform to alloy AG40a or AC41a of QQ-Z-363.

3.4.8 Epoxy resin. Epoxy resin shall be a stoning epoxy and have a minimum shore D hardness of 85 and incorporate a ultra violate inhibitor.

3.5 Design. The embossed design of the insignia shall be an exact replica of the design of the Government furnished hubs (see 3.9), from which the contractor's working dies shall be extracted. The working dies shall be tooled and polished to remove any dents, nicks, scratches, or other imperfections.

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3.6 Construction.

3.6.1 Stamping, trimming and piercing. The insignia shall be struck in a manner to insure a well defined die-struck edge. When applicable, insignia shall be trimmed and pierced to the die-struck outline. All edges shall be clean and smooth, free from burrs, drag, step and rough edges. The piercing and trimming operations, when applicable, shall not damage or distort the design or alter the shape of the insignia.

3.6.1.1 Die casting. When die cast method is applicable, the insignia shall be free from porosity, blow holes or other casting defects. Before finishing all flash lines shall be removed and the edges of the insignia shall be smooth. Insignia is die cast from zinc aluminum specified in para 3.4.7 and shall be copper plated. When tested as specified in paragraph 4.7.8 the plating shall be a minimum of 0.0003 inch thick.

3.6.2 Soldering and electronic fusion. Unless otherwise specified on the applicable specification sheet, all soldering shall be accomplished using hand solder specified in 3.4.2.1. Joints shall be clean, smooth, strong and free from burned or reduced areas. There shall be no excess solder and all excess flux shall be removed. When tested as specified in 4.7.3, the soldered parts shall not seperate. When tested as specified in 4.7.7, the prong shall not seperate at the joint. Where soft solder is specified, soft solder specified in 3.4.2.2 shall be used. When tested as specified in 4.7.4, the soldered device or prong shall not seperate from the insignia.

3.6.3 Attaching devices. Unless otherwise specified on the applicable specification sheet, all insignia shall have a prong and clutch type attaching device.

3.6.3.1 Prong and clutch type. Unless otherwise specified prongs shall be made from nickel silver specified in 3.4.1. The prongs as shown on drawing B-13-12 (figure 1) shall be 5/16 inch plus or minus 1.64 inch in length and shall be driven, swaged, soldered or electronically fused in the locations specified in the applicable specification sheet. Clutches as shown on drawing B-13-12 (figure 1) shall be made from any type brass. When tested as specified in 4.7.2.1, it shall not be possible to remove the clutch from the prong without first depressing the release wings.

3.6.3.1.1 Prong (registering). Prongs shall be made from nickel silver specified in 3.4.1. The prongs shall be of the dimensions shown on the applicable specification sheet. They shall be positioned and attached as required by the applicable specification sheet.

3.6.3.1.2 Prong (integral). When prongs are die cast as integral part of the insignia, and tested in accordance with

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paragraph 4.7.10, no evidence of breaking or cracking shall occur.

3.6.3.2 Screw post type and thumb nut. Screw post type and thumb nut attaching devices shall be made from brass specified in 3.4.1 and shall conform to the details and dimensions as shown in fugure 2. The screw post shall be made from free cutting brass and hard soldered or electronically fused to the back of the insignia in the location shown. When tested as specified in 4.7.2.2, thumb nuts shall not bind on the post's threads.

3.6.4 Forcers. When forcers are permitted to bring up the design of the insignia, the backs of the insignia shall have no sharp or rough edges. The impression of the forcer shall be symmetrical with the embossed design of the insignia. Sufficient metal shall be left near any trimmed area to assure clean smooth trimming without distortion.

3.7 Finish. The front and edges (surfaces visible from the front) shall be finished as specified on the applicable specification sheet and shall match the finish of the standard sample, the approved guide sample (see 3.2), or the approved first article (see 3.3). Where plating, anodizing or oxidizing is specified the entire item (front, back and edges) except for the attaching devices, shall be plated or oxidized as applicable. The finish on the backs of gold plated items shall match the standard sample. Unless otherwise specified on the applicable specification sheet the backs of all oxidized items shall be clean i.e., uniformly oxidized. In addition to the finish specified on the specification sheets all insignia not gold plated, anodized or subdued shall be thoroughly lacquered with clean lacquere specified in 3.4.3. Gold plated insignia may be lacquered at the option of the manufacturer. Except for aluminum insignia, all subdued insignia shall be given a black chemical finish conforming to MIL-F-495. The face and edges of the insignia shall then be coated with flat dead black enamel as specified in 3.4.6.2.1.

3.7.1 Cutting down. When necessary to remove nicks, scratches, inequalities, pores or other blemishes, the insignia shall be cut down by means of grease or oil buffing. The insignia shall not be cut down to the extent that any of the design is obliterated.

3.7.2 Plating.

3.7.2.1 Gold plating. Gold plating shall be accomplished using gold specified in 3.4.4. The use of white metal as an undercoating shall not be permitted. When tested as specified in 4.7.4, no visible chemical reaction such as evolution of gases shall appear.

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3.7.2.2.2 Silver plating. Silver plating shall be accomplished by electroplating methods using the silver specified in 3.4.5.2. Prior to plating, the surface to be plated shall be cleaned of all matter which would affect the plating. Brass items shall be hot nickel underplated a minimum of 0.00025 inch thick. Silver filled items shall not be underplated. The silver plating shall be a minimum of 0.0003 inch thick and shall be smooth, fine grained, adherent, free from visisble blisters, pits, porosity, indications of excessive build up or other defects. Nickel underplating thickness shall be tested in accordance with standard commercial practice. Items made of silver filled material shall be tested on the reverse side for compliance with the silver plating thickness requirements and items made of brass shall be silver thickness tested on the obverse side.

3.7.2.3 Rhodium plating. When tested as specified in 4.7.4 or 4.7.10 no chemcial reaction shall occur. When rhodium plating is required over brass, an undercoating of hot nickel shall be required.

3.7.2.4 Hot Nickel underplating. When nickel plating is specified on the applicable specification sheet, the underplating shall not be less than 0.00025 inch. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM B487.

3.7.3 Lacquering. Unless gold plated all items shall be lacquered using lacquer specified in 3.4.3, as applicable (see 3.7 and appropriate specification sheet). The dry lacquer film shall be continuous, level, adherent and free from lint, dust or other inclusions. When tested as specified in 4.7.5.1, there shall be no change in appearance of the lacquered surface except for a slight yellowing of the highlights and not more than two spots, each less than 1/16 inch largest dimension. When tested as specified in 4.7.5.2, the tissue paper shall not adhere to the lacquered surface.

3.7.4 Enameling.

3.7.4.1 Hard enameling. Hard enameling shall be accomplished using enamel specified in 3.4.6.1, and shall be charged, fired, stoned level with the dykes and lapped, felted or wheel polished to produce a hard glass like finish. The finish shall be adherent and free from bubbles, pits, foreign inclusions, cracking, crazing, burned edges or darkened emanel.

3.7.4.2 Baked enameling. Baked enameling shall be accomplished using enamel specified in 3.4.6.2. When tested as specified in 4.7.6.1, and 4.7.6.2, the baked enamel shall not flake or chip and only a slight residue of color shall be transferred to the cotton.

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3.7.5 Subdued insignia. The face and edges of all subdued insignia shall be coated with enamel specified in 3.4.6.2.1. Prior to enameling the entire insignia shall be given a black chemical finish conforming to MIL-F-495. When tested as specified in 4.7.6.2, only a slight amount of color shall be permitted to transfer to the cotton.

3.7.6 Anodizing. Aluminum insignia shall be anodized to a minimum depth of 0.0003 inch and dyed, when applicable, to the required color. All anodized insignia shall be sealed in a nickel acetate bath and when tested as specified in 4.7.11, no stain shall appear on the anodize.

3.8 Marking. The contractor shall stamp his trade or other identifying mark, the letter GI (GI for Government procurement only) and any fineness or quality marking necessary legibly and inconspicuously on the back of each item. Silver filled insignia may be marked in any of the following ways: 1/20 S.F., 1/20 Sil Filled, or 1/20 silver filled.

3.9 Government loaned property. Hubs will be loaned by the Government and shall be used to make the contractor's working dies necessary for one contract or order.

3.10 Workmanship. Insignia shall be clean, well made, and shall meet the acceptable quality levels established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either

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indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5 and 4.6 in this specification.

4.4 First article inspection. Inspection and testing of the first article shall be made on a completely finished item for all provisions of this specification applicable to end product examination and tests.

4.5 Quality conformance inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.6 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.6.1 Inspection of components.

4.6.1.1 Testing of components. In addition to the quality assurance provisions of the subsidiary specifications and drawings, inspection shall be performed on components and materials listed in table II for the test characteristics shown therein. The inspection level shall be S-1 and the requirements shall be applicable to the lot average.

TABLE II. Testing of components.

Components and lots expressed in terms of	Characteristic	Rqm't ref	Test method	#determ per s/u	Results reported as	Sample Unit
Copper base alloy (as applicable) (1 lb)	Chemical composition	3.4.1	4.7.1	2 ea element	Nearest 0.1% ea. element	4 ozs. material each alloy or 10 prongs when applicable

4.6.1.2 Testing of the intermediate product. Insignia shall be tested for copper plating thickness in accordance with 4.6.2.2.

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There shall be two determinations per sample unit and the results shall be reported to the nearest 0.0001 inch. The sample unit shall be one plated insignia (from each plating batch) of one type.

4.6.1.3 Certification of compliance. Components and materials listed herein may be accepted on the basis of a contractor's certification of compliance for the requirements indicated.

Component	Characteristic	Rqm't ref
Sterling Silver	Fineness	3.4.5.1
Silver for plating	Fineness	3.4.5.2
Gold for plating	Karat	3.4.4
Hard Solder	Material identification	3.4.2.1
Soft Solder	Material identification	3.4.2.2
Clear lacquer	Material identification	3.4.3
Hard enamel	Material identification	3.4.6.1
Soft enamel	Material identification	3.4.6.2
Gold plating	No white metal used as undercoating	3.7.2.1
Nickel silver for prong	Material identification	3.6.3.1
Brass for screw post and thumb nut	Material identification	3.6.3.2

4.6.2 Inspection of the end item.

4.6.2.1 Visual inspection of the insignia. The defects found during examination shall be classified in accordance with 4.6.2.1.1 and 4.6.2.1.2. The lots shall be inspected in accordance with 4.6.2.1.3. The unit of products for these examinations shall be one completely fabricated insignia.

4.6.2.1.1 Examination of insignia for defects in finish, design, material construction, workmanship and marking. Defects designated by an asterisk (*) shall be classified as major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously.

TABLE III Testing of the end item

Examine	Defect	Classification Major	(*)	Minor
Insignia (front)				
Color and finish	Color or finish of metal does not compare favorable with standard sample or approved first article	-	*	-
	Discoloration, such as spot, stain or speck	-	*	-
	Not clean	-	-	x

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Examine	Defect	Classification Major	(*)	Minor
	Paired insignia not matching in color, shade, and finish (when applicable)	-	*	-
	Any solder spatter clearly noticeable ...	-	*	-
	Oxidized finish has not been sufficiently relieved, or has been excessively relieved	-	*	-
	Background is not finished as specified..	x	-	-
Quality of metal (unplated items)	Surface is pitted, porous, crystalline, open grained or otherwise impaired	x	-	-
Enameling (soft)	Not enameled or area of no film	x	-	-
	Area of thin film	-	-	x
	Enamel not flat and dead black	x	-	-
	Not adherent, i.e., blistered, flaking or peeling	-	*	-
	Foreign matter imbedded	-	*	-
	Discoloration clearly noticeable	-	*	-
	Coating not continuous and level, i.e., runs, drips, or drops	-	*	-
Enameling (hard)	Not enameled (when required)	x	-	-
	Wrong color	x	-	-
	Cracked, chipped or crazed	-	*	-
	Not stoned smooth (when required) e.g., is coarse or uneven	-	*	-
	Is not level with dikes, e.g., contains high or low spots	-	*	-
	Hole or blister	-	*	-
	Laps, overruns or skips	-	*	-
	Not clean, e.g., blemish such as spot, stain, spatter or imbedded foreign matter	-	*	-
Lacquer (except gold plated items[3]	Lacquer coating hazy, cloudy or powdering.	-	-	x
	Foreign matter imbedded in finish, clearly noticeable	-	-	x
	Lacquer not smooth, not continuous or not adherent, i.e., flaking, blistering, rippling, peeling or has run, sag, or drops	-	*	-
	Lacquer not set to touch	-	-	x
Design	Construction or design details altered or obscured or not conforming to the Government hub	x	-	-
	Not dapped (when required)	x	-	-
	Any significant detail not clear, missing or obliterated, i.e., does not compare favorably with approved guide sample	x	-	-

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Examine	Defect	Classification Major	(*)	Minor
	Any significant detail marred or reduced..	-	*	-
	Any warp, twist or distortion, producing irregular surface contour or outline	-	*	-
	Any detail struck over resulting in a double impression	-	*	-
	Any area not pierced (when required)	-	*	-
Quality of plating	Any plated area cut through, not continuous, broken, peeling, blistered or otherwise impaired	-	*	-
	Not plated when required or plating is not specified type	x	-	-
	Foreign matter imbedded	-	*	-
Workmanship and construction	Surface or edge not clean, not smooth or free from burrs, roughness, drag, step or tool marks	-	*	-
	Metal marks on exposed surface such as nick, dent, dig, gouge or scratch	-	*	-
	Edge not die-struck, when applicable	x	-	-
	Not trimmed to the die-struck edge, when applicable	-	*	-
	Die-struck edge, rim or outline is not well defined, when applicable	x	-	-
	Pierced edge not clean and smooth e.g., any burr, drag, cutter, or file mark, when applicable	-	*	-
	Design or shape is damaged, distorted or altered by piercing operation, when applicable	x	-	-
	Forcer used when not specified or insignia forced in other than specified location	-	*	-
	Impression of the forcer (when applicable) is not symmetrical with the design on the front	-	-	x
	Metal torn or broken out by use of forcer.	-	-	x
	Any component missing, bent, twisted, broken, or malformed	-	*	-
	Solder does not completely unite parts, i.e., any perceptible opening between component parts	-	*	-
	Area burned or reduced in soldering	-	*	-
	Soldered joint not clean or smooth	-	-	x
	Excess solder or excess flux not removed .	-	-	x

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Examine	Defect	Classification Major (*)	Minor	
Insignia (back)				
Finish	Not clean, i.e., discoloration, such as spot or stain, over 3/16 inch in maximum dimension	-	-	x
	Not finished as specified	x	-	-
Attaching device (as applicable)	Defective, i.e., any part damaged or malformed which affects use or which may be expected to affect use	x	-	-
	Deformed, i.e., any part misshaped, bent, damaged, but not affecting use	-	-	x
	Not type specified	x	-	-
	Not positioned as specified	x	-	-
	Not attached in specified manner, e.g., not soldered or swaged as required	x	-	-
	Metal strip not clinched around shank of prong (when required)	x	-	-
	Clutch fastener will not engage or is loose fit on prong	x	-	-
	Swaged prong can be removed by hand	x	-	-
	Both swaged prongs are loose and wobble	-	-	x
	Threads on post or nut are stripped, crossed, damaged or not specified size or type	x	-	-
	Any component omitted, misplaced, operation omitted or not properly performed	-	*	-
	Clutch pierced through by prong	-	-	x
Identification marking	Missing, illegible, misspelled, incorrect, not stamped or not inconspicuously placed	-	-	x

4.6.2.1.2 Examination of insignia for defects in dimensions. Any dimension that is not within the specified tolerance shall be classified as a defect.

4.6.2.1.3 Inspection levels and AQL's. The inspection levels and the acceptable quality levels (AQL's) expressed in defects per hundred units (DHU) shall be as follows:

	Inspection level	AQL Major	AQL Total
For Defects applicable to 4.6.2.1.1	II	2.5	15.0
For Defects applicable to 4.6.2.1.2	S-3	(one class)	4.0

4.6.2.2 Testing of the end item. Testing of the completely fabricated insignia shall be performed in accordance with table III for the characteristics shown therein. The sample unit for

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the test for fused joints shall be one, the inspection level shall be S-2 and the AQL shall be 6.5. The sample unit for all other applicable end item tests shall be four and the inspection level shall be S-1 and the AQL shall be 6.5. Requirements shall be applicable to the individual unit. Regardless of the number of prong failures on a sample unit, it shall be classified as one defective unit.

TABLE III. Testing of end item.

Characteristic	Rqm't paragraph	Test method	#determ per s/u	Results reported as
Test for clutch fastener and prongs	3.6.3.1	4.7.2.1	1	Pass or fail
Integral prong test	3.6.3.1.2	4.7.9 1/	8	Pass or fail
Test for screw posts and thumb nuts	3.6.3.2	4.7.2.2	1	Pass or fail
Test for electronically	3.6.2	4.7.7	1	Pass or fail
Test for hand soldered joints	3.6.2	4.7.3	1	Pass or fail
Acid test for gold plate	3.7.2.1	4.7.4	1	Pass or fail
Silver plating thickness	3.7.2.2	Std com'l 2/	1	Pass or fail
Nickel underplating	3.7.2.4	Std com'l 3/	1	Pass or fail
Test for clear lacquer:				
Liver of sulfur	3.7.3	4.7.5.1	1	Pass or fail
Tackiness test	3.7.3	4.7.5.2	1	Pass or fail
Test for baked enamel:				
Knife test	3.7.4.1	4.7.6.1	1	Pass or fail
Acetone test	3.7.4.1	4.7.6.2	1	Pass or fail

1/ Insignia are to be tested per sample unit as two prongs per insignia.
2/ In the event of dispute, the test method shall be Test Method 52.1. of FED-SDT-151.
3/ In the event of dispute, the test shall be conducted in accordance with ASTM-B-487.

4.6.3 Examination for count of insignia in intermediate containers. Paired insignia and insignia of one design and finish packaged for shipment shall be examined to determine conformance with package markings and specified quantity. The sample unit for this examination shall be one box (intermediate container). The lot size shall be the number of intermediate containers. Any box containing less than the specified or marked quantity of insignia or pair of insignia

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mismated, when applicable shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

4.6.4 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements of section 5, of this specification are compiled with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred unit.

Examine	Defect
Marking (exterior) and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged affecting serviceability.
Workmanship	Inadequate application of components, such as, incomplete closure of container flaps, loose strapping or taping, inadequate stapling. Bulging or distortion of containers.
Content	Number of intermediate packages is more or less than required.

4.6.5 Palletization examination. An examination shall be made to determine that the palletization is in compliance with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the acceptable quality level (AQL), express in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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4.7 Test.

4.7.1 Chemical composition test. Chemical composition shall be determined in accordance with Test Method III of FED-STD-151. Results shall be evaluated to determine compliance with the requirements specified in 3.4.1.

4.7.2 Attaching device test.

4.7.2.1 Test for prong and clutch type. The clutches shall be removed and replaced 10 times from the prong using the clutch release wings. An attempt shall then be made to remove the clutch by hand without first pressing the release wings to determine compliance with the requirements specified in 3.6.3.1.

4.7.2.2 Test for screw post type and thumb nut. The thumb nut shall be assembled on the screw post, during which, the operation of screw threads shall be observed to determine compliance with the requirements specified in 3.6.3.2.

4.7.3 Test for hard soldered joints. Hard soldered insignia shall be placed in an oven maintained at 1065 degrees F., plus or minus 5 degrees F., for 15 minutes. While at this temperature the insignia shall be lifted by the attaching device and shall be inspected to determine compliance with the requirements specified in 3.6.2.

4.7.4 Test for soft soldered joints. Soft soldered insignia shall be placed in an oven maintained at 365 degrees F., plus or minus 5 degrees F., for 15 minutes. While at this temperature, the insignia shall be lifted by the superimposed device and shall be inspected to determine compliance with the requirements specified in 3.6.2.

4.7.5 Acid test for gold plating. The test acid shall be applied as follows: Place a drop of test acid (each drop shall be not less than 1/16 inch in diameter) on three different spots on the obverse of the plated surface allowing the drops to remain for not less than one minute, during which time the surface of the insignia shall be inspected to determine compliance with the requirements specified in 3.7.2.1 and 3.7.2.3. The test acid applied at room temperature shall consist of a solution containing 50 percent by volume of CP nitric acid (specific gravity 1.42) and an equal volume of distilled water. If the test insignia has been lacquered, the lacquer must be removed prior to testing.

4.7.6 Lacquer tests.

4.7.6.1 Liver of sulfur test. Insignia, except enameled Marine Corps or gold plated insignia shall be immersed in a 2

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percent by weight CP liver of sulfur (sulfated potash) water solution, at a temperature of 100 degrees F., plus or minus 10 degrees F., for a period of 3 minutes. The insignia shall then be removed and rinsed in warm, then cold, then hot water. Insignia shall be wiped gently with an absorbent cellulose material or whirled to remove residual moisture and shall be air dried at room temperature (60 to 80 degrees F.) for 1 hour. The insignia shall then be inspected to determine compliance with the requirements specified in 3.7.3.

4.7.6.2 Tackiness test. At room temperature (60 to 80 degrees F.) press a piece of tissue paper against the lacquered surface for 15 seconds, using any pressure capable of being exerted between thumb and two fingers, after which, the pressure shall be released and the insignia inspected to determine compliance with the requirements specified in 3.7.3.

4.7.7 Baked enamel tests.

4.7.7.1 Knife test. Draw a knife edge across the face side of the specimen marking four parallel scores 1/16 inch apart. Repeat this process drawing the knife this time perpendicular to the previously scored lines, making small blocks in the finish. The finish shall then be examined to determine compliance with 3.7.4.

4.7.7.2 Acetone test. The enameled surface shall be wiped five times with a piece of cotton saturated with acetone. The cotton shall be examined to determine compliance with 3.7.4.2.

4.7.8 Test for fused joints. Item to be tested shall be anchored on a horizontal surface. All prongs shall be grasped at least 1/3 its length above the fused joint and bent thru an angle of 90 degrees (45 degrees to each side of the vertical until seperation occurs). The fused joint shall then be examined to determine compliance with 3.6.2. A bending tool in the form of a 45 degree template may be used for this test provided the prong is gripped at least 1/3 its length above the fused joint.

4.7.9 Direct micro-thickness test for copper plating. Copper plating thickness shall be determined in accordance with method 521 of FED-STD-151. Results shall be evaluated to determine compliance with para. 3.6.1.1.

4.7.10 Test for integral prongs. Insignia to be tested will be placed in a jig which allows no movement of the insignia and positions prongs on a 30 degree angle above the horizontal. A vertical 10 lb. load shall be applied to the top third of each prong. The prongs will be examined to determine compliance with para. 3.6.3.1.2.

4.7.11 Continuity of rhodium plating. The unlacquered

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rhodium plated item shall be tested by immersion in a solution of liver of sulfur (2% sulfurated potash and 98% water, by weight, at a temperature of 100 degree F, ± 10 degree F) for a period of 1 minute. Upon removal the item shall be washed, cleaned, and examined for compliance with 3.7.2.3.

4.7.12 Dye stain test. A spot of phosphate-free anthorguinone violet RN dye shall be placed on the anodized surface. After five minutes the dye shall be wiped off and the surface examined to determine compliance with 3.7.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A, C or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. One pair of insignia shall be mounted on a white sulfite board card 2 1/2 inches square and not less than 0.011 inch thick or one each of Warrant Officer Candidate insignia, Officer Candidate insignia, cap or hat insignia, Air Force branch of service for Chaplain, Christian Faith, Chaplain, Jewish Faith, Nurse, Physician, Dentist or Judge Advocate General shall be mounted on a white sulfite board card 2 7/8 inches wide by 1 1/2 inches wide and not less than 0.011 inch thick. Each card shall then be packaged in a snug-fitting transparent polyethylene bag. The polyethylene film shall be a minimum of 0.002 inch thick. All seams and closures shall be affected by heat sealing. Prior to or during the heat sealing operation, excess air within the bag shall be expelled.

5.1.1.2 Intermediate packaging. Two hundred branch of service insignia or one hundred cap insignia, of one type only, shall be placed in a setup paperboard box conforming to type I, variety 1, class A, stype 4 of PPP-B-676; or in a folding paperboard box conforming to style III, type G, class 1 of PPP-B-566. Unit packages shall be stacked on end, face to face and back to back to fill the container. Inside dimensions of each size paperboard box required shall be determined by the contractor. Any void within the box shall be filled with crumpled tissue paper or other similar cushioning material to prevent the shifting of contents. Box closure shall be secured with 2-inch minimum width gummed paper tape conforming to type III of PPP-T-45 applied at the center of the length opening and extended along the bottom and up the sides at least 3 inches.

5.1.2 Level C. Insignia shall be preserved to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving station for immediate use or controlled humidity storage. The supplier may use his standard practice, when it meets these requirements.

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5.1.3 Commercial. Insignia shall be packaged in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be level A, B, or C or commercial as specified (see 6.2).

5.2.1 Level A. Insignia preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class weather resistant. Closure shall be in accordance with the appendix of the container specification of PPP-B-636. The weight of the contents of each shipping container shall not exceed the size and weight limitations of the container specification.

5.2.2 Level B. Insignia preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box conforming to class domestic, grade 275 of PPP-B-636. Closure shall be in accordance with appendix of the container specification. The weight of the contents of each shipping container shall not exceed the size and weight limitations of the container specification.

5.2.3 Level C. Insignia preserved as specified in 5.1, shall be packed in a manner to assure carrier acceptance and safe delivery at destination at the lowest transportation rates for such supplies. Containers shall be in accordance with the rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking for cleaning of gold plated items. Each mounted card for gold plated insignia shall have the following statement printed in bold pring on the face of the card: "To clean, rub gently with a soft dry cloth. Do not clean with a chemically treated cloth or abrasive."

5.3.1 Marking. In addition to any special marking required by the contract or order, unit packages, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM-D-3951, as applicable. Except unit package marking may be printed on the reverse of the mounting card provided the printing is continuous and is legible thru the bag.

5.4 Palletization. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. Insignia covered by this specification are intended to be worn by officer personnel of the Armed Forces to indicate the wearer is a member of a particular service, corps or branch.

6.2 Acquisition requirements.

6.2.1 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification

b. Title, number and date of the applicable specification sheet, (see 2.1 and 3.1)

c. When a standard sample or an approved guide sample will not be furnished (see 3.2).

d. When a first article is not required (see 3.3).

e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

f. When palletization is required (see 5.4).

6.2.2 Contract data requirements. Data conforming to Department of Defense Data Item Descriptions DI-R-4803, DI-T-4901, DI-T-4902, DI-T-4903 and DI-T-4904 will usually be required for delivery in connection with this specification for Marine Corps procurements. For other than Marine Corps procurement, the cognizant activity will provide comparable data item descriptions. When so required, such data will be specified for delivery on a DD-1423 included in the contract.

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.9.

6.5 First article. When a first article inspection is required, the item should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one completed insignia. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations

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for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Subject term key(word) listing.

Branch of Service Insignia
Garrison Cap and Service Cap
Officer personnel

6.7 Change from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of changes.

Custodians:
Army - IH
Navy - NU
Air Force - 11

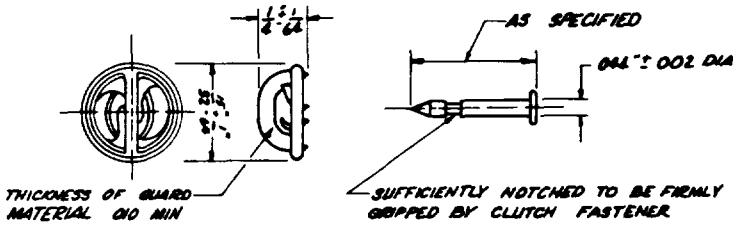
Review Activities:
Navy - MC
Air Force - 82, 99

User Activities:
Air Force - 45

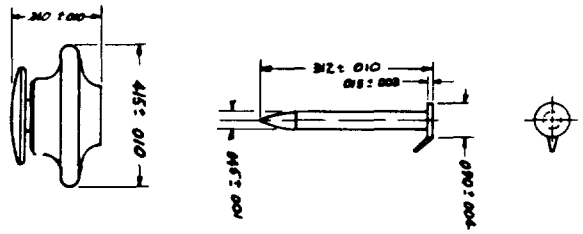
Preparing Activity:
Army - IH

(Project No. 8455-0308)

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CLUTCH PRONG



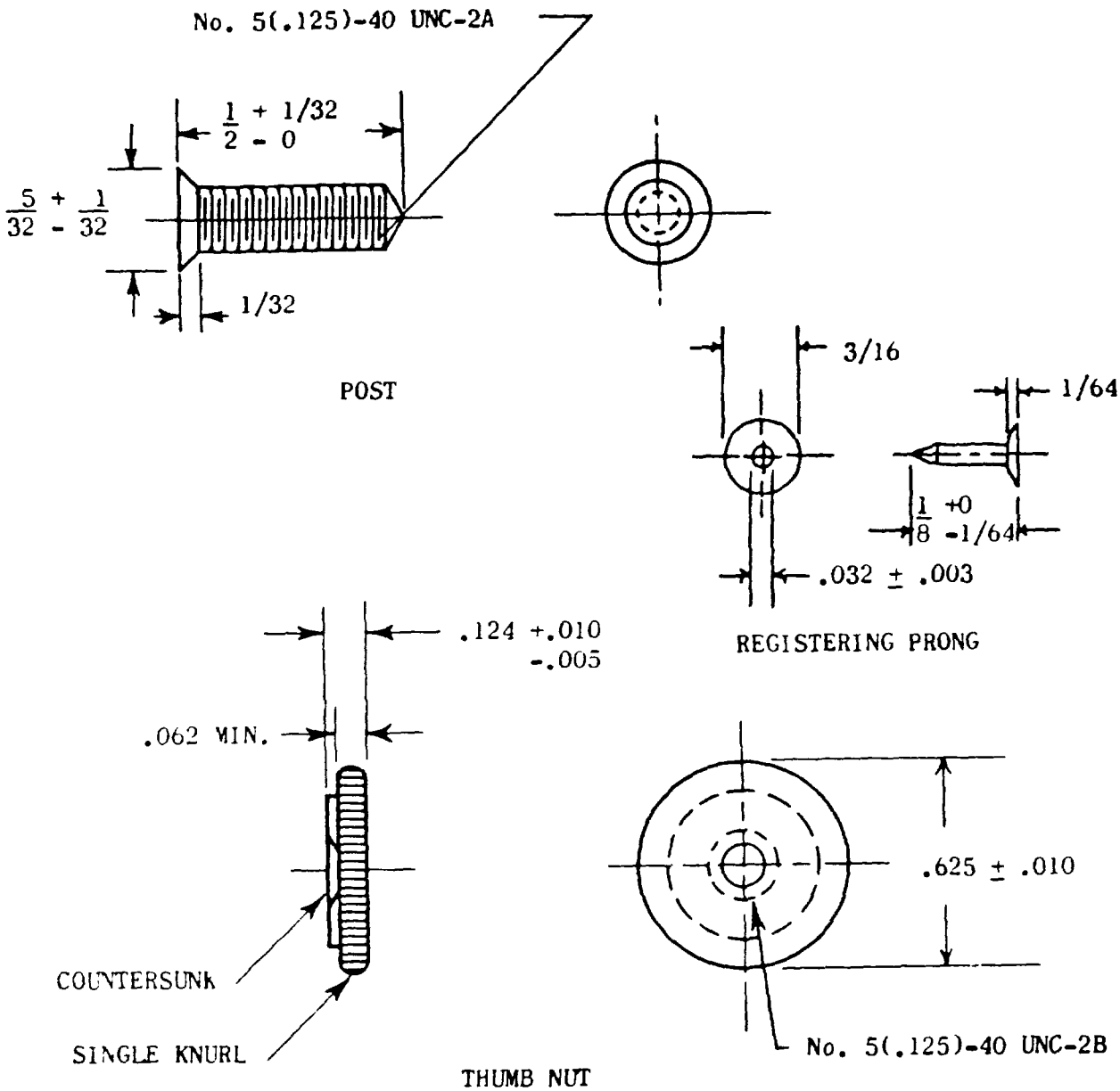
FLAT BALL TYPE CLUTCH PRONG WITH PICK

ATTACHING DEVICES HERALDIC
PRONG / CLUTCH TYPE

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FIGURE 1

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NOTE: Dimensions are in inches.

FIGURE 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-I-3575F	Insignia, Branch of Service, General Spec for

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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